1. Aim of the practical:

a)WAP to demonstrate dialog boxes in JavaScript.
b) WAP to demonstrate cookies in JavaScript.
c) WAP to create form to submit user records and validate data.

2. Objective:

- To develop a user-friendly web form that collects user information, ensuring that the data entered meets specified criteria.
- To implement form validation techniques to ensure data integrity and provide immediate feedback to users.
- To understand the importance of client-side validation in improving user experience and reducing server-side errors.

3. Code

A)WAP to demonstrate dialog boxes in JavaScript.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Confirm and Prompt Example</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin: 0;
            padding: 0;
            color: white;
            height: 100vh;
            display: flex;
            flex-direction: column;
            justify-content: center;
            align-items: center;
            overflow: hidden;
            animation: changeColor 5s infinite;
        }

        h1 {
            font-size: 2.5em;
            margin-bottom: 20px;
            text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.5);
        }

        button {
            padding: 15px 30px;
            font-size: 1.2em;
```

```
    cursor: pointer;
    border: none;
    border-radius: 5px;
    background-color: #ff6b6b;
    color: white;
    transition: background-color 0.3s ease;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
}

button:hover {
    background-color: #ff4c4c;
}

.moving-text {
    position: absolute;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%);
    font-size: 3em;
    white-space: nowrap;
    pointer-events: none;
    display: none;
    animation: move 2s linear infinite, changeColor 5s infinite, rotate 2s linear infinite;
}

@keyframes changeColor {
    0% { color: #ff6b6b; background-color: #153807; }
    25% { color: #ff9b6b; background-color: #1917b6; }
    50% { color: #ff6b9b; background-color: #2e0730; }
    75% { color: #9b6bff; background-color: #112331; }
    100% { color: #6bff9b; background-color: #4caf50; }
}

@keyframes rotate {
    0% { transform: translate(-50%, -50%) rotate(0deg); }
    100% { transform: translate(-50%, -50%) rotate(360deg); }
}
```

```
        @media (max-width: 600px) {
            h1 {
                font-size: 2em;
            }

            button {
                font-size: 1em;
                padding: 10px 20px;
            }

            .moving-text {
                font-size: 2em;
            }
        }
    </style>
</head>
<body>
    <h1>Eligible for Voting or Not</h1>
    <button id="greetButton">Check!!</button>
    <div id="movingText" class="moving-text">You are Eligible!</div>

    <script>
        document.getElementById("greetButton").addEventListener("click", function() {
            let age = prompt("Please enter your Age:");

            if (age && !isNaN(age) && age.trim() !== '') {
                age = parseInt(age);

                if (age > 0) {
                    let confirmAge = confirm(`Would you like to know if you are eligible for
voting?`);

                    if (confirmAge) {
                        if (age >= 18) {
                            alert(`Yay, you are eligible!`);
                            const movingText = document.getElementById("movingText");
                            movingText.style.display = "block"; // Show the moving text
                        } else {
```

```
                    alert(`No bud, you need to wait ${18 - age} more years!`);
                }
            }
        } else {
            alert("Age should be a positive number!");
        }
    } else {
        alert("Please enter a valid number for your age.");
    }
});
</script>
</body>
</html>
```

B) WAP to demonstrate cookies in JavaScript.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Cookie Demonstration</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            padding: 20px;
        }

        label, input, button {
            display: block;
            margin: 10px 0;
        }

        #message {
            margin-top: 20px;
            font-weight: bold;
```

```
        }

        #cookieList {
            margin-top: 20px;
            font-size: 0.9em;
            color: #555;
        }
    </style>
</head>
<body>
    <h1>Cookie Demo</h1>

    <label for="cookieName">Cookie Name:</label>
    <input type="text" id="cookieName" placeholder="Name">

    <label for="cookieValue">Cookie Value:</label>
    <input type="text" id="cookieValue" placeholder="Value">

    <label for="cookieDays">Days to Expire:</label>
    <input type="number" id="cookieDays" placeholder="Days" min="0">

    <button id="setCookieBtn">Set Cookie</button>
    <button id="getCookieBtn">Get Cookie</button>
    <button id="deleteCookieBtn">Delete Cookie</button>

    <div id="message"></div>

    <div id="cookieList"></div>

    <script>
        function setCookie(name, value, days) {
            let expires = days ? "; expires=" + new Date(Date.now() + days * 864e5).toUTCString()
: "";
            document.cookie = name + "=" + (value || "") + expires + "; path=/";
            displayAllCookies();
        }

        function getCookie(name) {
```

```
        return document.cookie.split('; ').reduce((acc, c) => {
            const [key, val] = c.split('=');
            return key === name ? decodeURIComponent(val) : acc;
        }, null);
    }

    function deleteCookie(name) {
        document.cookie = name + '=; Max-Age=-99999999; path=/';
        displayAllCookies();
    }

    function displayAllCookies() {
        const cookies = document.cookie.split("; ").map(cookie =>
cookie.split("=")).map(([key, value]) => `${key}: ${value}`).join("<br>");
        document.getElementById('cookieList').innerHTML = cookies ? `<h3>All
Cookies:</h3><p>${cookies}</p>` : "<h3>No Cookies Set</h3>";
    }

    document.getElementById('setCookieBtn').onclick = () => {
        const name = document.getElementById('cookieName').value.trim();
        const value = document.getElementById('cookieValue').value.trim();
        const days = document.getElementById('cookieDays').value.trim();

        if (!name || !value) {
            document.getElementById('message').textContent = "Please provide both cookie
name and value.";
            return;
        }

        setCookie(name, value, days ? parseInt(days) : 0);
        document.getElementById('message').textContent = `Cookie "${name}" set!`;
        clearInputs();
    };

    document.getElementById('getCookieBtn').onclick = () => {
        const name = document.getElementById('cookieName').value.trim();
        if (!name) {
```

```
            document.getElementById('message').textContent = "Please provide a cookie name to
retrieve.";
            return;
        }

        const value = getCookie(name);
        document.getElementById('message').textContent = value ? `Cookie "${name}":
${value}` : `Cookie "${name}" not found!`;
    };

    document.getElementById('deleteCookieBtn').onclick = () => {
        const name = document.getElementById('cookieName').value.trim();
        if (!name) {
            document.getElementById('message').textContent = "Please provide a cookie name to
delete.";
            return;
        }

        deleteCookie(name);
        document.getElementById('message').textContent = `Cookie "${name}" deleted!`;
        clearInputs();
    };

    function clearInputs() {
        document.getElementById('cookieName').value = '';
        document.getElementById('cookieValue').value = '';
        document.getElementById('cookieDays').value = '';
    }

    displayAllCookies();
  </script>
</body>
</html>
```

C)WAP to create form to submit user records and validate data.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>User Records Form</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            background-color: #f0f0f0;
        }

        .container {
            background-color: white;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
            width: 300px;
        }

        h2 {
            text-align: center;
            margin-bottom: 20px;
        }

        label {
            display: block;
            margin-bottom: 5px;
```

```
        }

        input {
            width: 100%;
            padding: 8px;
            margin-bottom: 10px;
            border: 1px solid #ccc;
            border-radius: 4px;
        }

        .error {
            color: red;
            font-size: 0.8em;
            margin-bottom: 10px;
        }

        button {
            width: 100%;
            padding: 10px;
            background-color: #4caf50;
            color: white;
            border: none;
            border-radius: 4px;
            cursor: pointer;
        }

        button:hover {
            background-color: #45a049;
        }
    </style>
</head>
<body>

<div class="container">
    <h2>Submit User Record</h2>

    <form id="userForm">
        <div>
```

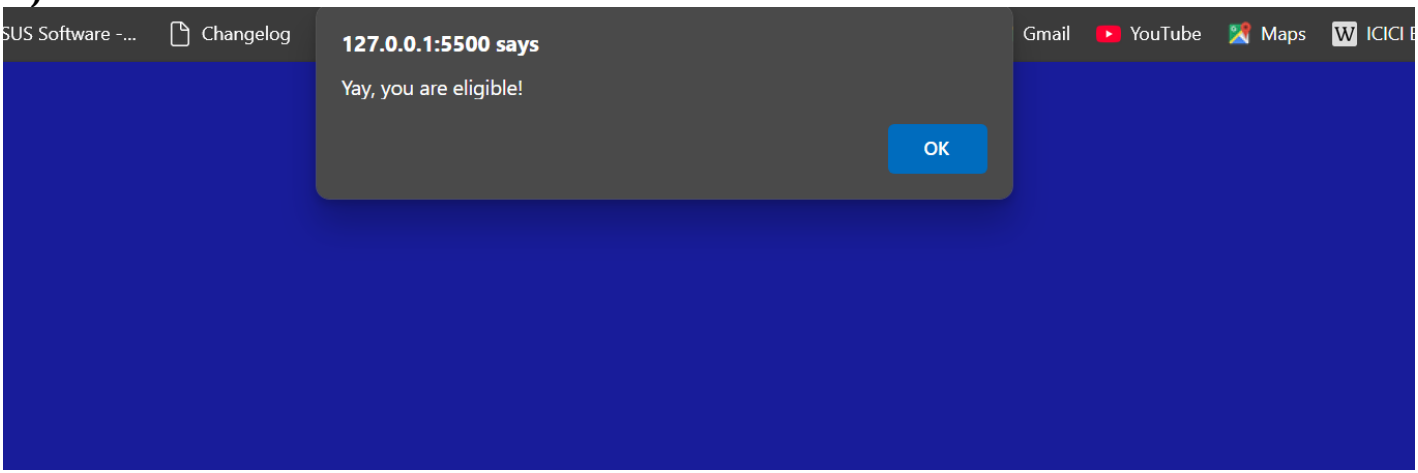
```
        <label for="name">Name:</label>
        <input type="text" id="name" name="name" placeholder="Enter your name">
        <div class="error" id="nameError"></div>
    </div>

    <div>
        <label for="email">Email:</label>
        <input type="text" id="email" name="email" placeholder="Enter your email">
        <div class="error" id="emailError"></div>
    </div>

    <div>
        <label for="password">Password:</label>
        <input type="password" id="password" name="password" placeholder="Enter your
password">
        <div class="error" id="passwordError"></div>
    </div>

    <div>
        <label for="age">Age:</label>
        <input type="number" id="age" name="age" placeholder="Enter your age">
        <div class="error" id="ageError"></div>
    </div>

    <button type="submit">Submit</button>
  </form>
</div>

<script>
  document.getElementById("userForm").addEventListener("submit", function(event) {
    event.preventDefault();

    // Clear previous error messages
    document.getElementById("nameError").textContent = "";
    document.getElementById("emailError").textContent = "";
    document.getElementById("passwordError").textContent = "";
    document.getElementById("ageError").textContent = "";
```

```
    // Fetch form values
    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let password = document.getElementById("password").value.trim();
    let age = document.getElementById("age").value.trim();

    let isValid = true;

    // Name validation
    if (name === "") {
        document.getElementById("nameError").textContent = "Name is required.";
        isValid = false;
    }

    // Email validation (simple pattern)
    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
    if (email === "") {
        document.getElementById("emailError").textContent = "Email is required.";
        isValid = false;
    } else if (!emailPattern.test(email)) {
        document.getElementById("emailError").textContent = "Invalid email format.";
        isValid = false;
    }

    // Password validation (at least 6 characters)
    if (password === "") {
        document.getElementById("passwordError").textContent = "Password is required.";
        isValid = false;
    } else if (password.length < 6) {
        document.getElementById("passwordError").textContent = "Password must be at least 6
characters.";
        isValid = false;
    }

    // Age validation (must be 18 or older)
    if (age === "") {
        document.getElementById("ageError").textContent = "Age is required.";
        isValid = false;
```

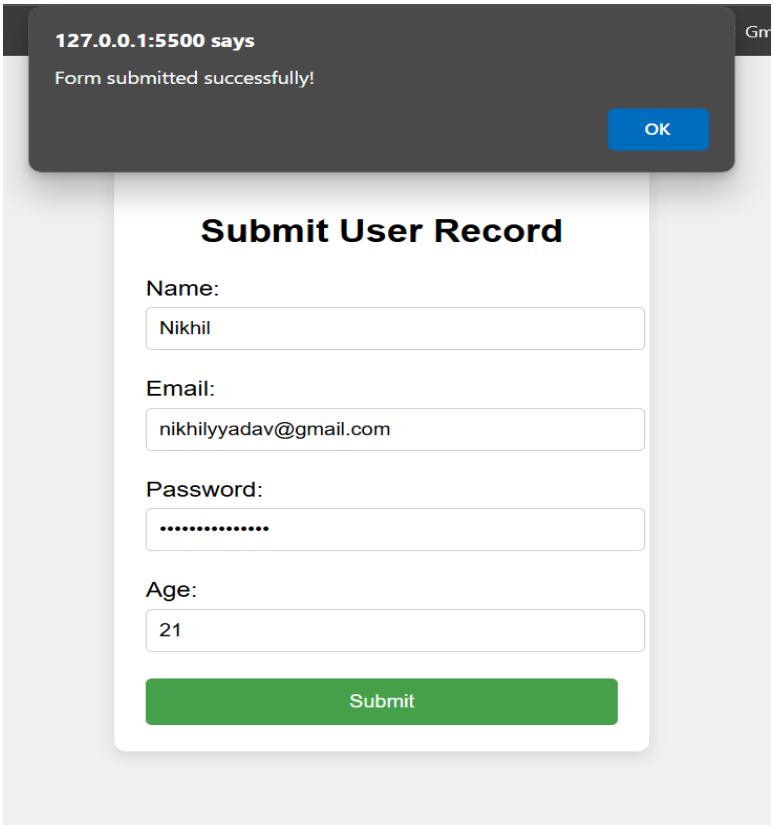
```
        } else if (isNaN(age) || age < 18) {
            document.getElementById("ageError").textContent = "You must be 18 or older.";
            isValid = false;
        }

        // If all validations pass, alert success
        if (isValid) {
            alert("Form submitted successfully!");
            // Here, you can also send the form data to a server or perform further actions.
        }
    });
</script>

</body>
</html>
```

4. Output

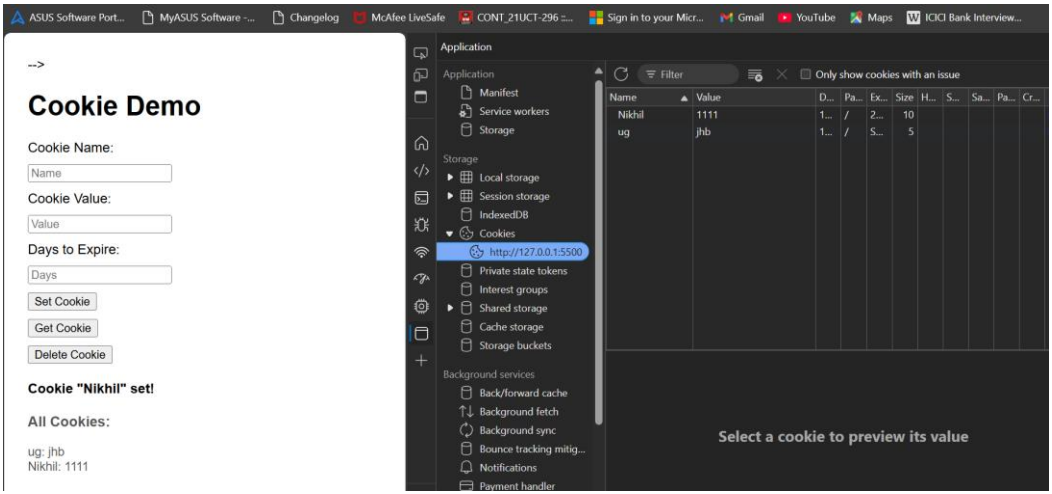
A)



B)


127.0.0.1:5500 says
Form submitted successfully!
OK
Submit User Record
Name:
Nikhil
Email:
nikhilyyadav@gmail.com
Password:
Age:
21
Submit

C)


Cookie Demo
Cookie Name:
Name
Cookie Value:
Value
Days to Expire:
Days
Set Cookie
Get Cookie
Delete Cookie
Cookie "Nikhil" set!
All Cookies:
ug: jhb
Nikhil: 1111
Application
Manifest
Service workers
Storage
Local storage
Session storage
IndexedDB
Cookies
http://127.0.0.1:5500
Private state tokens
Interest groups
Shared storage
Cache storage
Storage buckets
Background services
Back/forward cache
Background fetch
Background sync
Notifications
Payment handler
Only show cookies with an issue
Name
Value
Nikhil
1111
ug
jhb
Select a cookie to preview its value

5. Learning outcomes (What I have learned):

- Gain hands-on experience in managing cookies using JavaScript for client-side storage.
- Reinforce JavaScript skills through functions, event handling, and DOM manipulation.
- Learn to enhance user interaction with feedback mechanisms in web application

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet		8 Marks
2.	Viva		10 Marks
3.	Simulation		12 Marks
	Total		30 Marks

Teacher Signature